Vansh Sharma

AI/ML Engineer|Generative AI Engineer|FullStack AI Developer

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PROFESSIONAL SUMMARY

AI/MLEngineer and Full-Stack AI Developer with experience building Generative AI, Agentic AI, RAG, enterprise automation, and data-intensive applications. Experienced in LLM integrations, multi-agent workflows, vector search, document intelligence, enterprise integrations, cloud-native backend systems, and Physical AI. Currently developing DAFFY at UBIQ Technologies using Rust, Polars, Computer Vision,IoT sensor data,aerial data, and industry-specific intelligence workspaces.

WORK EXPERIENCE

UBIQTech / DaffyOS,Noida

Jul 2026 – Present

Role: Full Stack Developer

- Developing DAFFYOS,an AI-driven business and industrial intelligence ecosystem that unifies organizational operations, data, and workflows through a centralized intelligence layer and industry-specific workspaces.
- Building high-performance data processing and analytics workflows using Rust and Polars for structured, operational, and sensor-generated datasets.
- Contributing to Physical AI systems connecting software intelligence with real-world environments through IoT sensors, aerial data, computer vision, and physical-world signals.
- Developing industry-specific workspaces that contextualize operational data and intelligence around business roles, workflows, and domain requirements.
- Working with computer vision and aerial imagery/data to support intelligent perception and analysis of physical environments and industrial operations.
- Processing and integrating IoT sensor data to connect physical assets and environments with digital intelligence, analytics, and decision-support workflows.
- Developing scalable backend and data-intensive components with focus on performance, reliability, efficient processing, testing, debugging, and deployment.

LiteratiTech Solutions Pvt Ltd / BotIntelli, Noida

Dec 2025 – Jul 2026

Role:FoundingSoftwareEngineer – AI/ML&FullStack

- Led end-to-end development of BotIntelli, an AI-powered enterprise automation platform.
- Designed and implemented multi-agent AI workflows using LLMs and orchestration frameworks.
- Built Retrieval-Augmented Generation (RAG) pipelines using vector embeddings and semantic search for enterprise knowledge retrieval.
- Integrated 10+ LLM providers, including OpenAI, Gemini, Ollama, and custom models, into production workflows.
- Built ingestion pipelines supporting 20+ enterprise data sources, including CRM systems, cloud storage, databases, and APIs.
- Engineered document intelligence pipelines using OCR, embeddings, chunking strategies, and vector retrieval.
- Integrated Salesforce, HubSpot, Zendesk, Google Analytics, Jira, and custom APIs for enterprise workflows.
- Designed scalable backend services using Python, FastAPI, Flask, Node.js, and MongoDB; implemented vector search architecture with Milvus.
- Built AI insights dashboards, workflow monitoring, execution analytics, and customer-facing automation tools.
- Managed Linux deployments using Nginx, PM2, Docker, cloud infrastructure, and CI/CD pipelines; collaborated with founders on product strategy and architecture.

Butterfly Technology LLC, New Delhi

Mar 2025 – Jun 2025

Role:SoftwareDeveloper – Flutter / Node.js / Python / OCR / Jira / Git

- Developeda full-stack Expense Managementapplicationusing Flutter and Node.js/Express for expense tracking, categorization, and visualization.
- Built RESTful backend APIs and automation scripts supporting mobile application workflows and data processing.
- Integrated EasyOCR and PDFPlumber for automated extraction from receipts and invoices.
- Integrated Google Gemini to interpret and validate extracted receipt and document data.
- Implemented interactive analytics and visualizations for budget insights and spending trends.

PROJECTS

DAFFYOS—Physical AI & Industrial Intelligence Ecosystem

Tech: Rust, Polars, Computer Vision, IoT, Aerial Data, Physical AI

- AI-driven ecosystem connecting organizational operations and industry-specific workspaces through a unified intelligence layer.
- High-performance data processing using Rust and Polars across operational, structured, and sensor-generated data.
- Physical AI workflows combining IoT sensor data, aerial imagery/data, computer vision, and physical-world signals for industrial intelligence.
- Context-aware workspaces designed around industry requirements, operational workflows, and authorized business information.

BotIntelli — Enterprise AI Automation Platform

Tech:Python,FastAPI,Flask,Node.js,Next.js,React,MongoDB,Milvus, OpenAI, Gemini, LangChain, Ollama, AWS S3, Apache NiFi, Linux, Docker, Nginx

- Developed an enterprise AI workflowautomation platform with no-code workflow capabilities,AI agents,and automation pipelines.
- Implemented multi-model architecture supporting OpenAI, Gemini, Claude, and local LLMs.
- Built vector search, document intelligence, enterprise RAG, CRM/ticketing integrations, analytics, and reporting modules.
- Designed scalable architecture andcontributed to implementation, testing, debugging, deployment, and enterprise delivery.

Expense Management Application

Tech:Flutter,Dart,Node.js,Express.js,EasyOCR, PDFPlumber, Google Gemini

- Developed a cross-platform expense management application for Android and iOS with expense tracking, categorization, and analytics.
- Implemented Bloc state management and RESTful APIs for clean architecture and predictable application state.
- Automated receipt/invoice extraction using OCR and PDF processing, with Gemini-based interpretation and validation.

Vamsi — AI Voice Assistant

Tech:Python,Pyttsx3,SpeechRecognition, Wikipedia, BeautifulSoup, Tkinter, YouTube Transcript API, Pygame

- Built a modular AI-poweredvoice assistant integratingspeech-to-text, text-to-speech, websearch,Wikipediaqueries, alarms, and productivity utilities.
- Recognized for three consecutive years as College Project of the Year for innovative AI assistant design.

TECHNICAL SKILLS

AI / Machine Learning: Generative AI, Agentic AI, LLMs, Prompt Engineering, AI Agents, Multi-Agent Systems, RAG, Semantic Search, Embeddings, Fine-Tuning Concepts, AI Automation, Vector Databases, AI Workflows, Physical AI

LLM & AI Frameworks: LangChain, LangGraph, OpenAI API, Gemini API, Ollama, Hugging Face, CrewAI, MCP, Function Calling, Tool Calling

Programming: Python, JavaScript, TypeScript, Rust, Dart

Frontend: Next.js, React.js

Backend Development: FastAPI, Flask, Django, Node.js, Express.js, REST APIs, WebSockets, Microservices, High-Performance Backend Development

Databases & Storage: MongoDB, PostgreSQL, MySQL, Milvus, Pinecone, AWS S3

Cloud & DevOps: AWS, GCP, Linux, Docker, Nginx, CI/CD, GitHub Actions, PM2, Cloud Storage

Data Engineering: ETL Pipelines, Data Ingestion, Data Transformation, Polars, Apache NiFi, API Integrations, Data Warehousing, Large-Scale Data Processing

Computer Vision & Spatial Intelligence: Computer Vision, Aerial Imagery, Aerial Data Processing, Physical Environment Analysis, Visual Data Processing

IoT & Physical Systems: IoT Sensors, Sensor Data Processing, Physical-World Data Integration, Real-World Operational Data

Enterprise Integrations: Salesforce, HubSpot, Zendesk, GA4, Jira, OAuth2, SSO, CRM Integrations

OCR & DocumentAI:EasyOCR,TesseractOCR,PDFPlumber, Google Vision API, Document Intelligence

CERTIFICATIONS & COURSES

- Software Architecture & Design of Modern Large Scale Systems
- Software Architecture & System Design – Practical Case Studies
- Python for Data Science and Machine Learning

EDUCATION

Bachelor of Technology (B.Tech) — Computer Science Engineering 2021 – 2025
Maharishi Dayanand University, Rohtak,Haryana

Higher Secondary Certificate (HSC) — Computer Science 2020 – 2021
Mother India Senior Secondary School(CBSE), Jhajjar,Haryana

Secondary School Certificate (SSC) 2018 – 2019
SFDAV Public School (CBSE), Mansurpur,Uttar Pradesh

ACHIEVEMENTS

- Google Developer Event Participant (2x): Selected and invited to attend Google Developer Events twice.
- College Project of the Year — Vamsi AI Assistant: Recognized for three consecutive years for an innovative AI-powered voice assistant.
- Sports Coordinator of the Year — IITM College of Engineering: Recognized for leadership and team management as College Sports Coordinator and Volleyball Team Captain.